

Tutorial 3 + Lecture 10 / Module 1

DC Circuit Analysis and Theorems

06/08/2026 07/08/2026

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Summary of Module 1

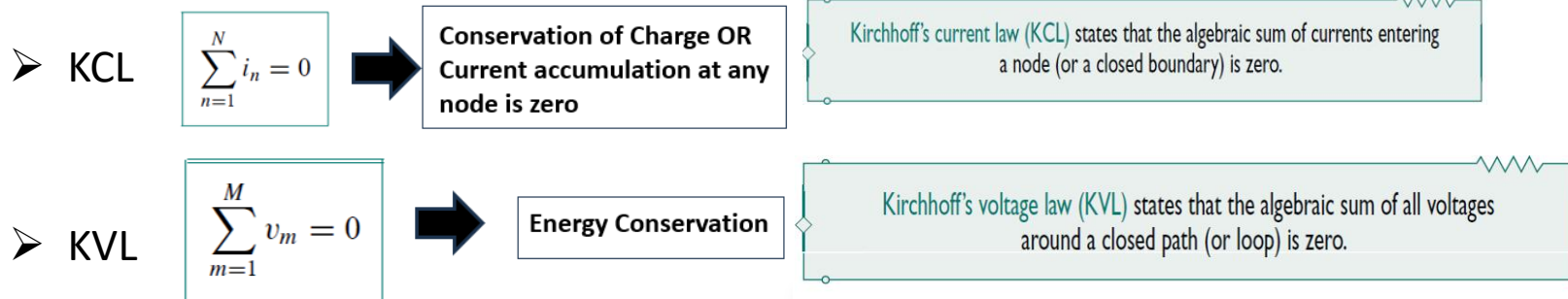
Section 1: Circuit, Network, Nodes, Branches, Loops, Meshes & Circuit Reduction

- Circuit is a closed path formed by electrical elements, Network is an interconnection of multiple electrical elements, **Circuit is a subset of Network**. Planar, Non-Planar Circuit
- **Active Elements** (Voltage Source, Current Source, Amplifier), **Passive Elements** (Resistors, Capacitors, Inductors), **Linear Elements** (Resistors, Capacitors, Inductors), Non-Linear Elements (Transistors)
- Circuit Analysis and Synthesis. DC and ac Analysis

$$b = l + n - 1$$

- b = Nos. of Branch, l = Nos. of Independent Loop/Mesh, n = Nos. of Node in a circuit
- We do simplify a complicated circuit by identifying the series/parallel connected resistors and then replacing them with equivalent resistance.
- When resistors are neither series nor parallel connected, they may be connected in T/Y/Star or Δ (Delta) configuration. In that case, simplify the circuit by applying the formula for T to Δ Or T to Δ

Section 2: KCL, KVL, Node and Mesh Analysis



- **Node Analysis** : To determine Node Voltages, Apply KCL, Independent and Dependent **Current Sources** , Node Analysis **with voltage source**: **Case I**: Voltage source connected between a non-reference and a reference node Consider voltage source value to be corresponding node voltage. **Case II**. Voltage Source connected between two non-reference nodes. Apply the concept of **Super Node** and follow the **prescribed steps** (Apply both KCL and KVL) for analysis.
- **Mesh Analysis** : To determine Branch Current, Apply KVL, Independent and Dependent **Voltage Sources** , Node Analysis **with Current source**: **Case I**: Current source is only within a single mesh. Consider the current source value as the corresponding branch current **Case II**. Current Source shared between two meshes. Apply the concept of Super Mesh and follow the **prescribed steps** (Apply both KCL and KVL) for analysis.

Section 3: Voltage Division, Current Division

- Voltage division applies on voltage source and series connected resistors
- Source voltage is divided among the resistors in direct proportion of their resistances. Larger the resistance larger the voltage drop. This is called principle of voltage division and the circuit is called voltage divider. In general, if a voltage divider has N resistors ($R_1, R_2, \dots, R_N$) in series with source voltage v , the n_{th} resistor (R_n) will have a voltage drop of

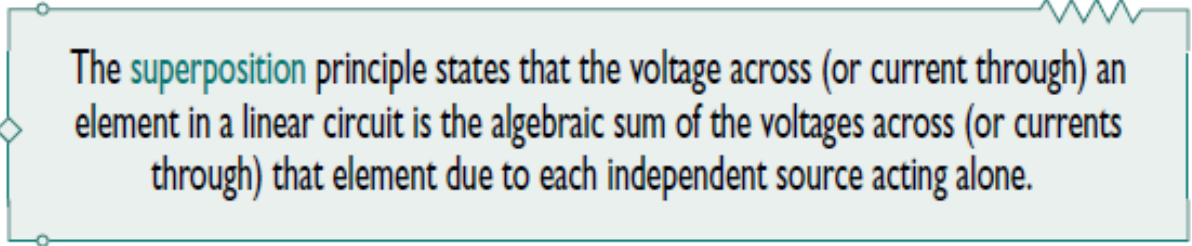
$$v_n = \frac{R_n}{R_1 + R_2 + \dots + R_N} v$$

- Current division applies on current source and parallel connected resistors
- Source current is divided among the resistors in inverse proportion of their resistances. Larger the resistance smaller the current flow. This is called principle of current division and the circuit is called current divider. In general, if a current divider has N resistors ($R_1, R_2, \dots, R_N$) in parallel with source current i , the n_{th} resistor (R_n) will have current flow of

$$i_n = \frac{1/R_n}{1/R_1 + 1/R_2 + \dots + 1/R_n} i$$

Section 4: Circuit Theorems

➤ Superposition theorem



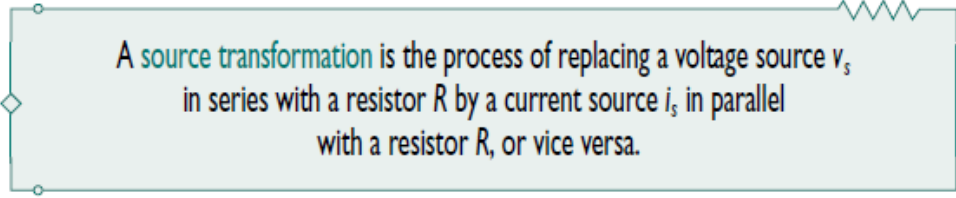
The **superposition** principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

Steps to Apply Superposition Principle:

1. Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using nodal or mesh analysis.
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

- One should practice problems on Superposition theorem for a circuit with **independent source alone (case I)** and with **both dependent and independent source (Case II)**. For Case II, while applying superposition theorem, **turn off all independent sources except one and leave the dependent source as it is** in Step 1 mentioned above.

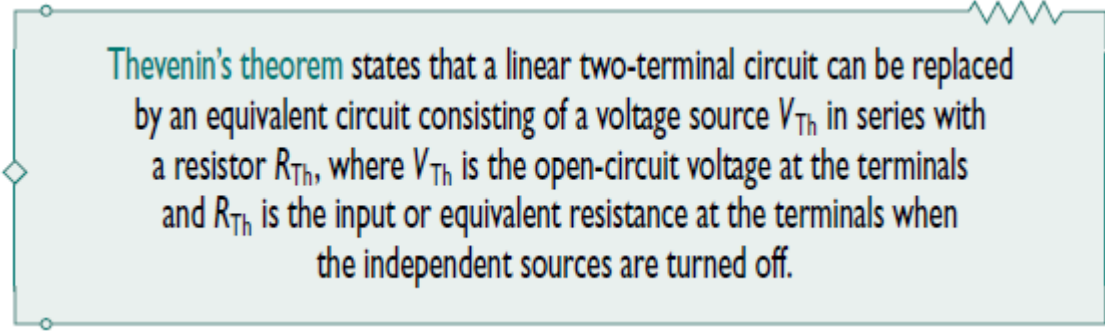
➤ Source Transformation



A **source transformation** is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.

- One should practice problems on Source Transformation for a circuit with **independent source alone (case I)** and with **both dependent and independent source (Case II)**.
- While doing source transformation the end nodes of voltage/current source and series/parallel resistor combined will remain same i.e.; the branches Connected to the nodes will be the same.

➤ Thevenin's Theorem



Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

- One should practice problems on Thevenin's Theorem for a circuit with **independent source alone (case I)** and with **both dependent and independent source (Case II)**.

➤ Thevenin's Theorem

$V_{Th} = v_{oc}$ (4.6)

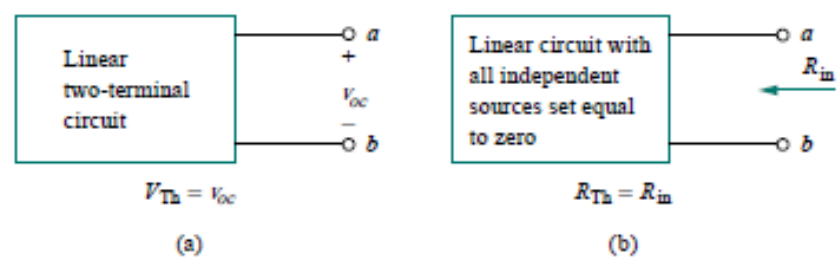


Figure 4.24 Finding V_{Th} and R_{Th} .

Again, with the load disconnected and terminals a - b open-circuited, we turn off all independent sources. The input resistance (or equivalent resistance) of the dead circuit at the terminals a - b in Fig. 4.23(a) must be equal to R_{Th} in Fig. 4.23(b) because the two circuits are equivalent. Thus, R_{Th} is the input resistance at the terminals when the independent sources are turned off, as shown in Fig. 4.24(b); that is,

$R_{Th} = R_{in}$ (4.7)

To apply this idea in finding the Thevenin resistance R_{Th} , we need to consider two cases.

CASE 1 If the network has no dependent sources, we turn off all independent sources. R_{Th} is the input resistance of the network looking between terminals a and b , as shown in Fig. 4.24(b).

CASE 2 If the network has dependent sources, we turn off all independent sources. As with superposition, dependent sources are not to be turned off because they are controlled by circuit variables. We apply a voltage source v_o at terminals a and b and determine the resulting current i_o . Then $R_{Th} = v_o/i_o$, as shown in Fig. 4.25(a). Alternatively, we may insert a current source i_o at terminals a - b as shown in Fig. 4.25(b) and find the terminal voltage v_o . Again $R_{Th} = v_o/i_o$. Either of the two approaches will give the same result. In either approach we may assume any value of v_o and i_o . For example, we may use $v_o = 1$ V or $i_o = 1$ A, or even use unspecified values of v_o or i_o .

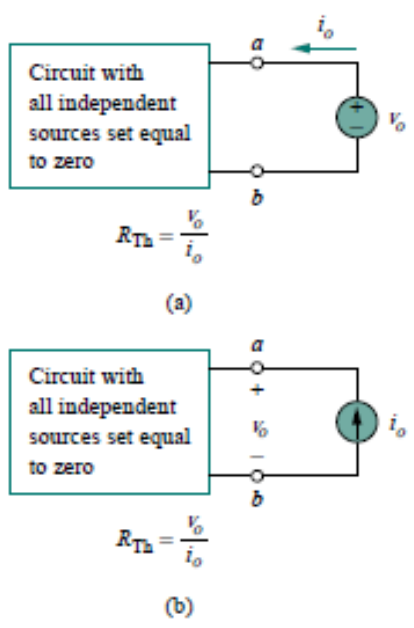


Figure 4.25 Finding R_{Th} when circuit has dependent sources.

Later we will see that an alternative way of finding

➤ Norton's Theorem

Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

$$R_N = R_{Th}$$

$$I_N = \frac{V_{Th}}{R_{Th}}$$

$$\begin{aligned} V_{Th} &= v_{oc} \\ I_N &= i_{sc} \\ R_{Th} &= \frac{v_{oc}}{i_{sc}} = R_N \end{aligned}$$

- One should practice problems on Thevenin's Theorem for a circuit with **independent source alone (case I)** and with **both dependent and independent source (Case II)**.

➤ Maximum Power Transfer Theorem

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).

The Thevenin equivalent is useful in finding the maximum power a linear circuit can deliver to a load. We assume that we can adjust the load resistance R_L . If the entire circuit is replaced by its Thevenin equivalent except for the load, as shown in Fig. 4.48, the power delivered to the load is

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \quad (4.21)$$

$$R_L = R_{Th}$$

$$p_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

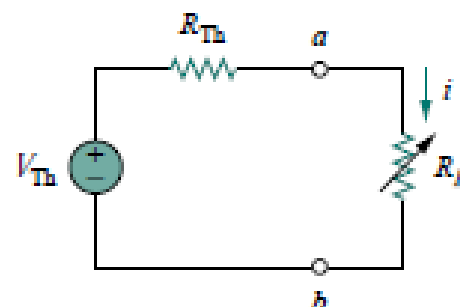
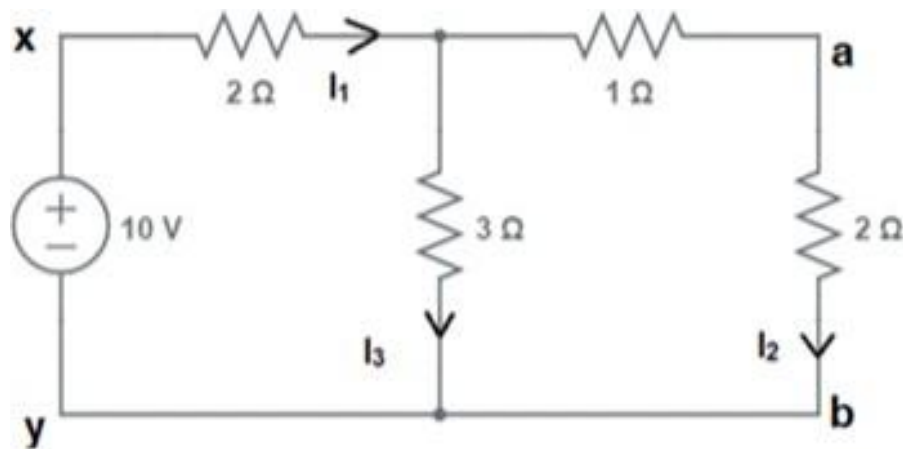


Figure 4.48 The circuit used for maximum power transfer.

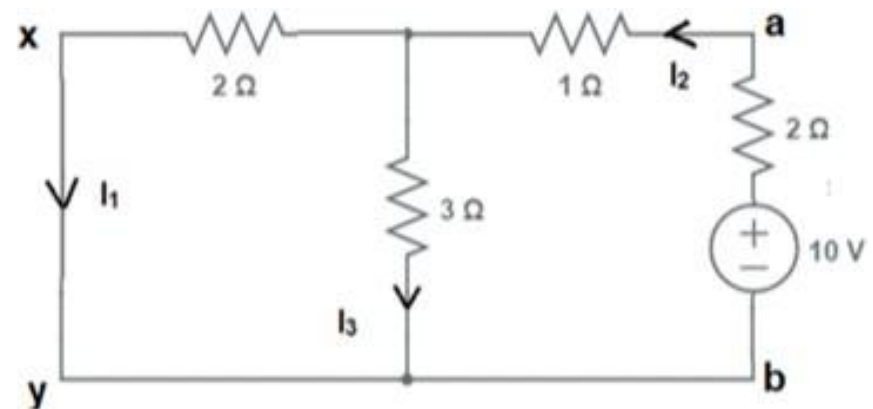
➤ Reciprocity Theorem

- In a linear, bilateral network containing only one independent source, if a voltage source V applied in branch A produces a current I in branch B, then placing the same voltage source V in branch B will produce the same current I in branch A.
- If a current source I placed in branch A produces a voltage V across branch B, then placing the same current source I in branch B will produce the same voltage V across branch A.



$$I_1 = 1.4 \text{ A}$$

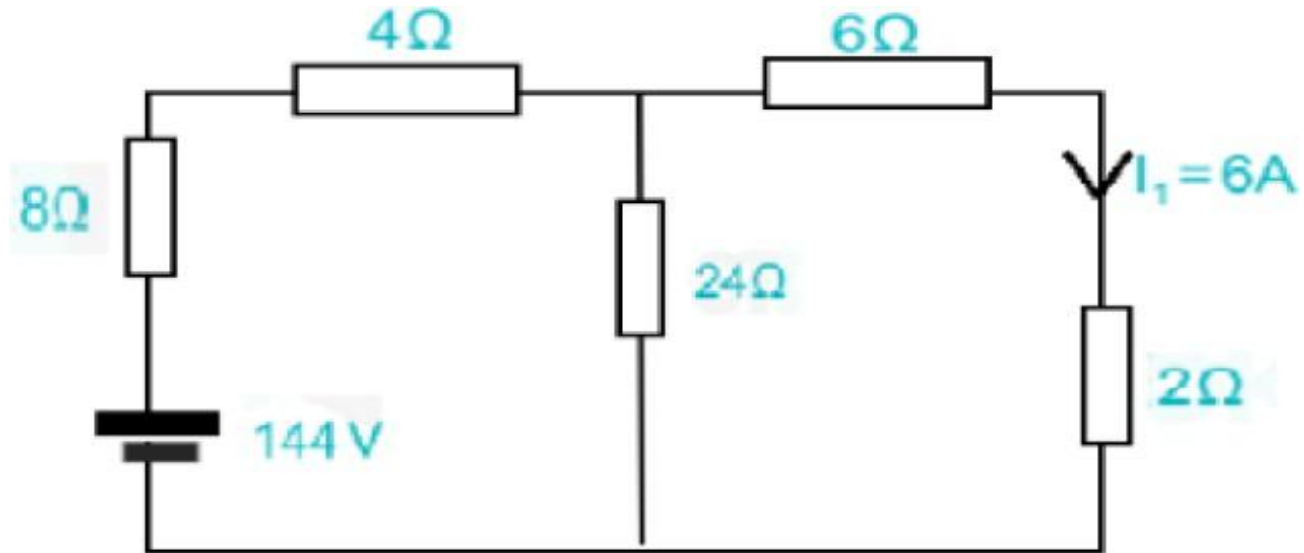
$$I_2 = 1.4 \text{ A}$$



Tutorial 3 Module 1

Problem 1

Validate the reciprocity theorem.



(a)

Problem 2

Find i_0 in the circuit using Superposition Theorem

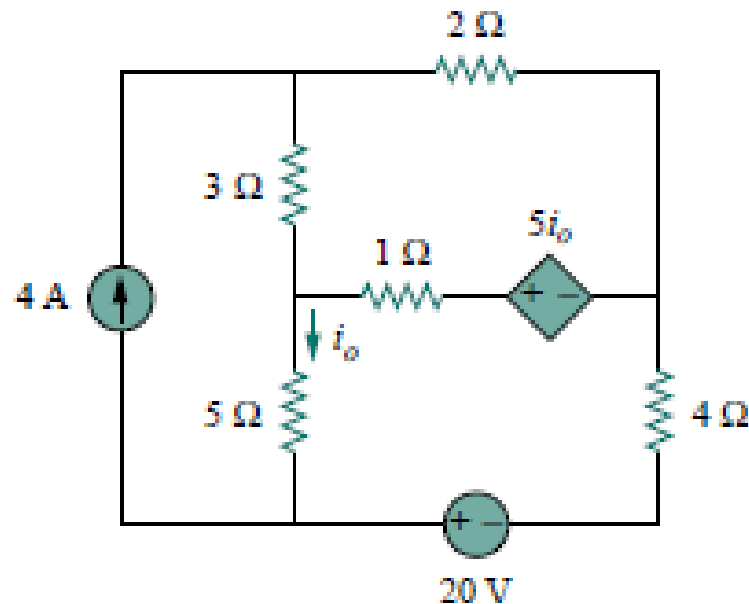


Figure 4.9 For Example 4.4.

Problem 2 Solution

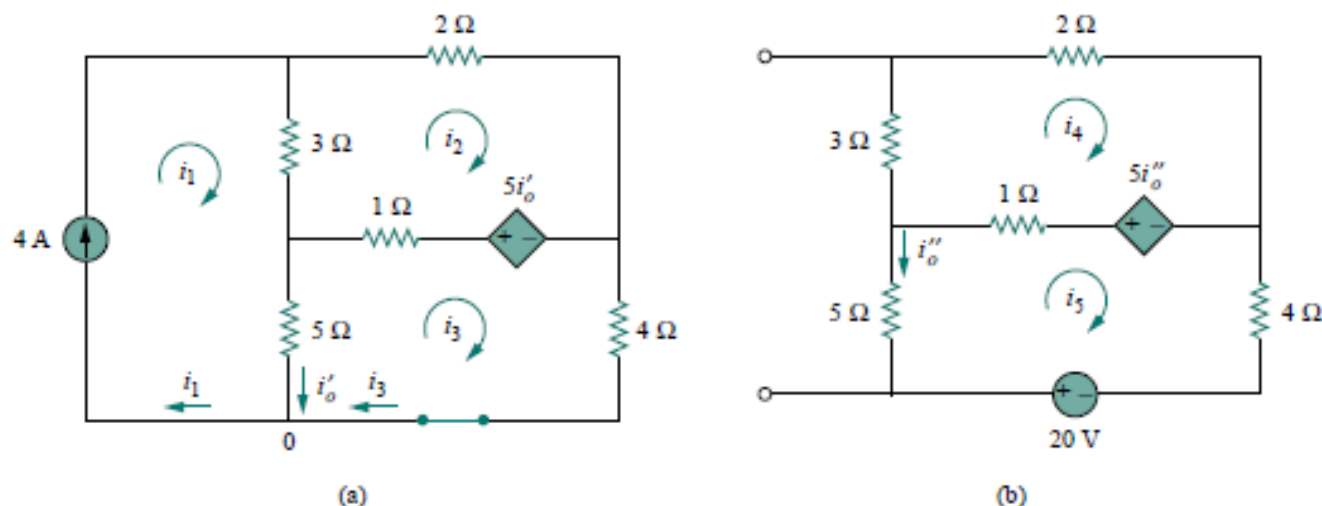


Figure 4.10 For Example 4.4: Applying superposition to (a) obtain i_o' , (b) obtain i_o'' .

The circuit in Fig. 4.9 involves a dependent source, which must be left intact. We let

$$i_o = i_o' + i_o'' \quad (4.4.1)$$

where i_o' and i_o'' are due to the 4-A current source and 20-V voltage source respectively. To obtain i_o' , we turn off the 20-V source so that we have the circuit in Fig. 4.10(a). We apply mesh analysis in order to obtain i_o' . For loop 1,

$$i_1 = 4 \text{ A} \quad (4.4.2)$$

For loop 2,

$$-3i_1 + 6i_2 - 1i_3 - 5i_o' = 0 \quad (4.4.3)$$

For loop 3,

$$-5i_1 - 1i_2 + 10i_3 + 5i_o' = 0 \quad (4.4.4)$$

For loop 3,

$$-5i_1 - 1i_2 + 10i_3 + 5i_o' = 0 \quad (4.4.4)$$

But at node 0,

$$i_3 = i_1 - i_o' = 4 - i_o' \quad (4.4.5)$$

Substituting Eqs. (4.4.2) and (4.4.5) into Eqs. (4.4.3) and (4.4.4) gives two simultaneous equations

$$3i_2 - 2i_o' = 8 \quad (4.4.6)$$

$$i_2 + 5i_o' = 20 \quad (4.4.7)$$

which can be solved to get

$$i_o' = \frac{52}{17} \text{ A} \quad (4.4.8)$$

To obtain i_o'' , we turn off the 4-A current source so that the circuit becomes that shown in Fig. 4.10(b). For loop 4, KVL gives

$$6i_4 - i_5 - 5i_o'' = 0 \quad (4.4.9)$$

Problem 3 Solution Continue

and for loop 5,

$$-i_4 + 10i_5 - 20 + 5i_o'' = 0 \quad (4.4.10)$$

But $i_5 = -i_o''$. Substituting this in Eqs. (4.4.9) and (4.4.10) gives

$$6i_4 - 4i_o'' = 0 \quad (4.4.11)$$

$$i_4 + 5i_o'' = -20 \quad (4.4.12)$$

which we solve to get

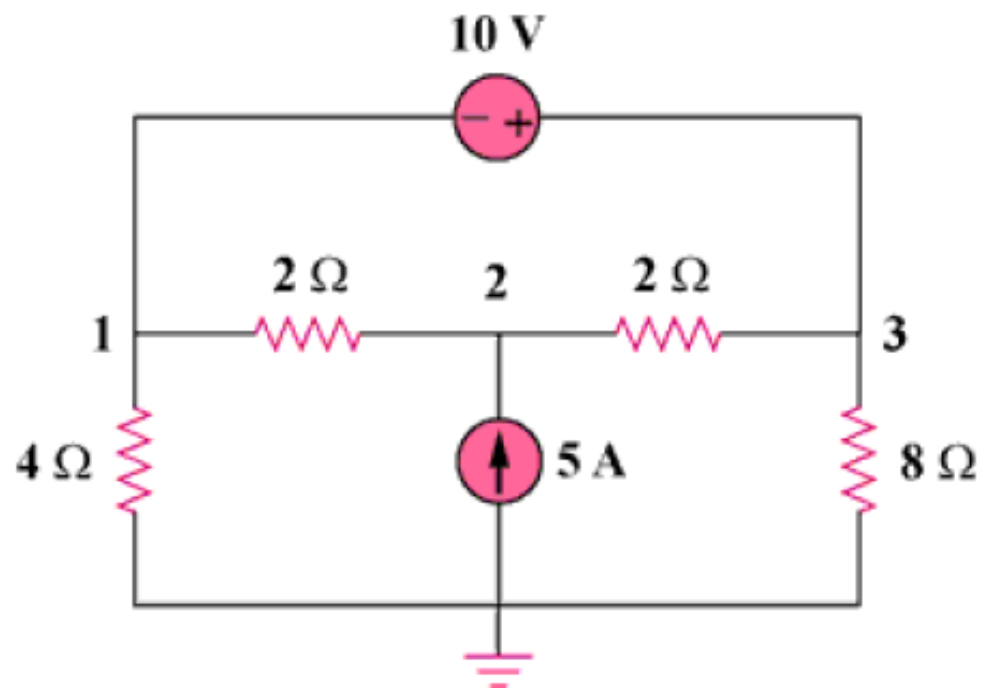
$$i_o'' = -\frac{60}{17} \text{ A} \quad (4.4.13)$$

Now substituting Eqs. (4.4.8) and (4.4.13) into Eq. (4.4.1) gives

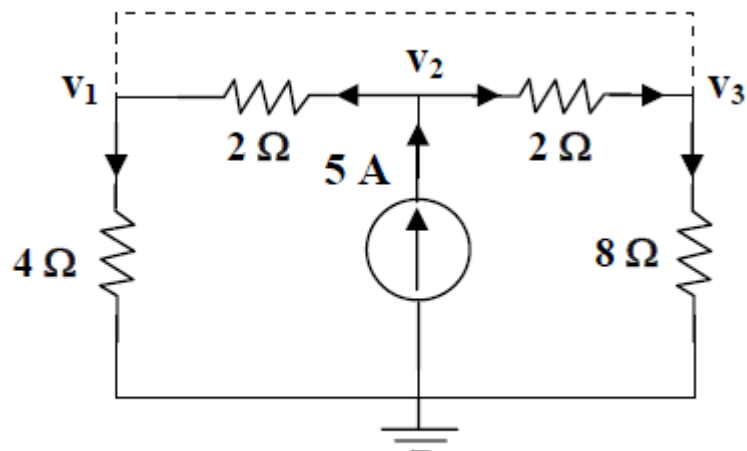
$$i_o = -\frac{8}{17} = -0.4706 \text{ A}$$

Problem 3

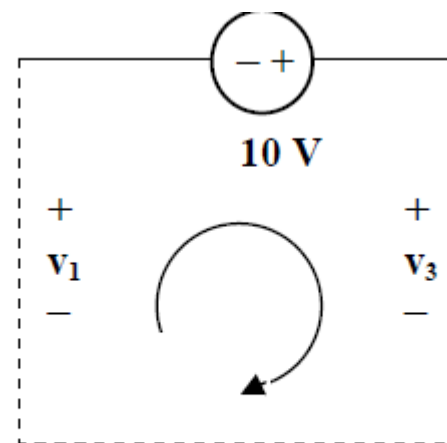
Determine the node voltages in the circuit in Fig. 3.67 using nodal analysis.



Problem 3 Solution



(a)



(b)

$$\text{At node 2, in Fig. (a), } 5 = \frac{v_2 - v_1}{2} + \frac{v_2 - v_3}{2} \longrightarrow 10 = -v_1 + 2v_2 - v_3 \quad (1)$$

$$\text{At the supernode, } \frac{v_2 - v_1}{2} + \frac{v_2 - v_3}{2} = \frac{v_1}{4} + \frac{v_3}{8} \longrightarrow 40 = 2v_1 + v_3 \quad (2)$$

$$\text{From Fig. (b), } -v_1 - 10 + v_3 = 0 \longrightarrow v_3 = v_1 + 10 \quad (3)$$

Solving (1) to (3), we obtain $v_1 = \underline{10 \text{ V}}$, $v_2 = \underline{20 \text{ V}} = v_3$

Problem 4

Determine the maximum power that can be delivered to the variable resistor R in the circuit of Fig. 4.139.

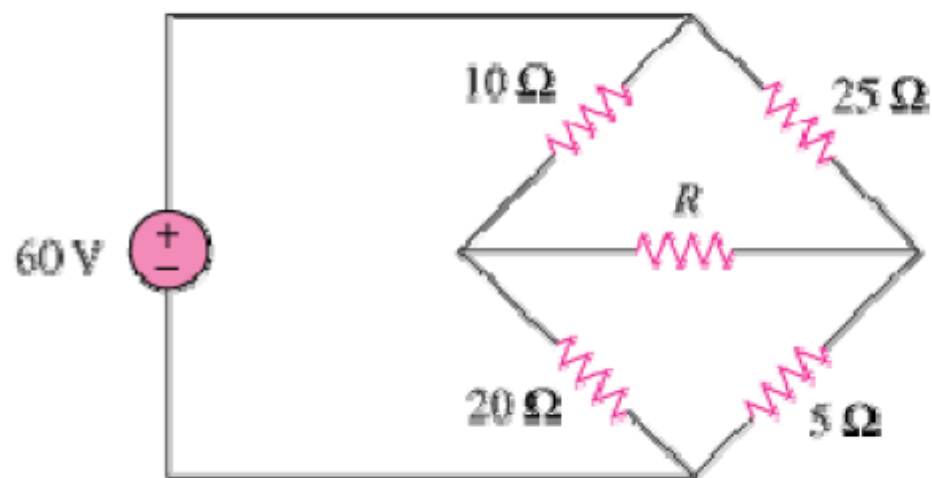
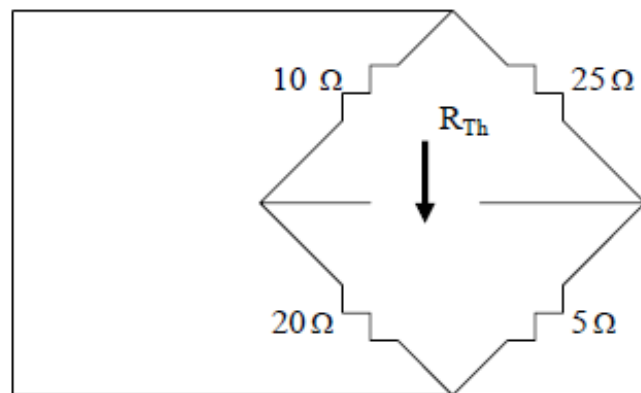


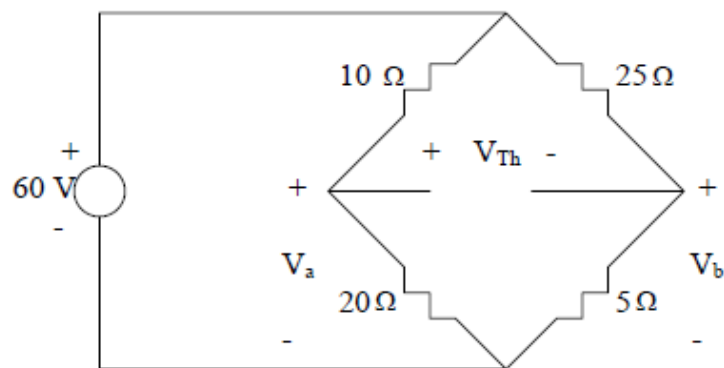
Figure 4.139

Problem 4 Solution

Find the Thevenin's equivalent circuit across the terminals of I



$$R_{Th} = 10 // 20 + 25 // 5 = 325 / 30 = 10.833 \Omega$$



$$V_a = \frac{20}{30}(60) = 40, \quad V_b = \frac{5}{30}(60) = 10$$

$$-V_a + V_{Th} + V_b = 0 \quad \longrightarrow \quad V_{Th} = V_a - V_b = 40 - 10 = 30 \text{ V}$$

$$P_{\max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{30^2}{4 \times 10.833} = \underline{20.77 \text{ W}}$$

Problem 5

For the circuit in Fig. 4.137, what resistor connected across terminals a - b will absorb maximum power from the circuit? What is that power?

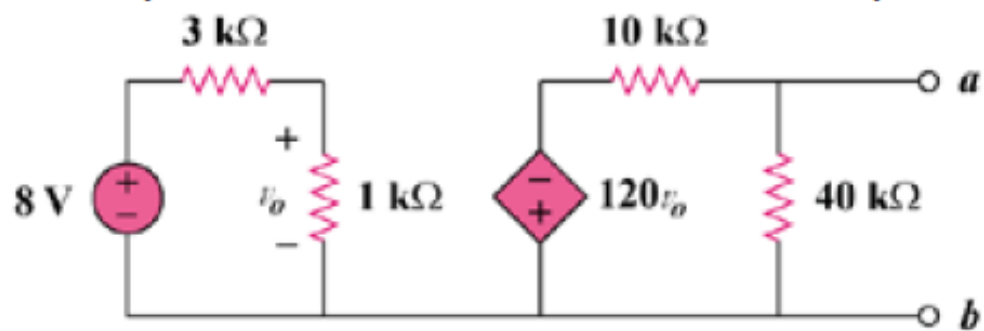
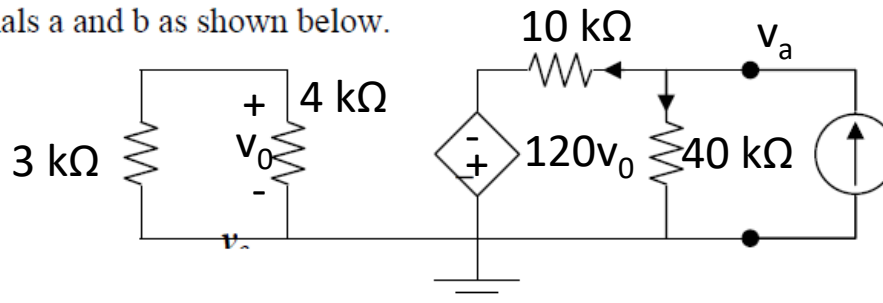


Figure 4.137

Problem 5 Solution

We need R_{Th} and V_{Th} at terminals a and b. To find R_{Th} , we insert a 1-mA source at the terminals a and b as shown below.



Assume that all resistances are in k ohms, all currents are in mA, and all voltages are in volts. At node a,

$$1 = (v_a/40) + [(v_a + 120v_o)/10], \text{ or } 40 = 5v_a + 480v_o \quad (1)$$

The loop on the left side has no voltage source. Hence, $v_o = 0$. From (1), $v_a = 8$ V.

$$R_{Th} = v_a/1 \text{ mA} = 8 \text{ kohms}$$

To get V_{Th} , consider the original circuit. For the left loop,

$$v_o = (1/4)8 = 2 \text{ V}$$

For the right loop, $v_R = V_{Th} = (40/50)(-120v_o) = -192$

The resistance at the required resistor is

$$R = R_{Th} = \underline{\underline{8 \text{ kohms}}}$$

$$p = V_{Th}^2/(4R_{Th}) = (-192)^2/(4 \times 8 \times 10^3) = \underline{\underline{1.152 \text{ watts}}}$$

Problem 6

Using Norton's theorem, find R_N and I_N of the circuit in Fig. 4.43 at terminals a - b .

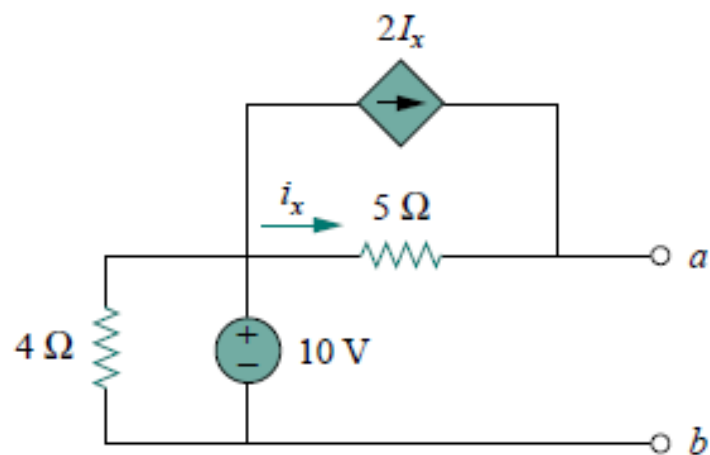
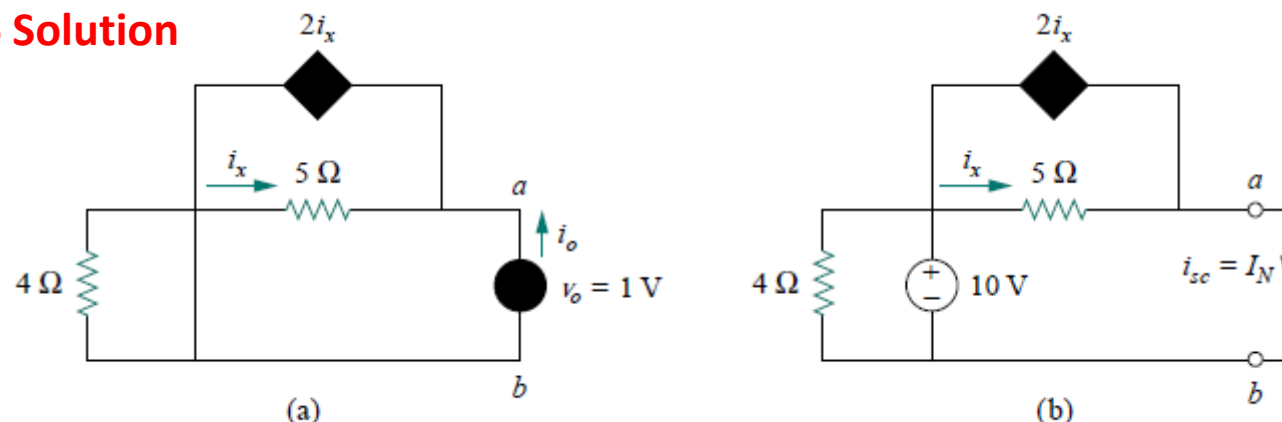


Figure 4.43 For Example 4.12.

Problem 6 Solution



terminals. We obtain the circuit in Fig. 4.44(a). We ignore the $4\text{-}\Omega$ resistor because it is short-circuited. Also due to the short circuit, the $5\text{-}\Omega$ resistor, the voltage source, and the dependent current source are all in parallel. Hence, $i_x = v_o/5 = 1/5 = 0.2$. At node a , $-i_o = i_x + 2i_x = 3i_x = 0.6$, and

$$R_N = \frac{v_o}{i_o} = \frac{1}{-0.6} = -1.67\ \Omega$$

To find I_N , we short-circuit terminals a and b and find the current i_{sc} , as indicated in Fig. 4.44(b). Note from this figure that the $4\text{-}\Omega$ resistor, the 10-V voltage source, the $5\text{-}\Omega$ resistor, and the dependent current source are all in parallel. Hence,

$$i_x = \frac{10 - 0}{5} = 2\text{ A}$$

At node a , KCL gives

$$i_{sc} = i_x + 2i_x = 2 + 4 = 6\text{ A}$$

Thus,

$$I_N = 6\text{ A}$$

Thank You